Sourcing as a Functor of Directed Containers: Routed Bills of Materials

Extracted from a machine-checked development

Abstract

We formalise the sourcing assignment of a bill of materials (BOM) as a covariant functor of directed containers. The domain is the directed container of *routed* BOMs: finite product-labelled rooted trees whose edges carry inbound shipping routes. The codomain is the shipping network read in the opposite direction—positions at a facility are inbound paths ending there. On shapes, the functor reads off the facility producing the root product; on positions, an occurrence is sent to the inbound route composed leg-by-leg through intermediate producers. Functoriality is *consolidation*: the empty occurrence is sourced by the empty route, and nested occurrences concatenate their routes. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

A bill of materials is a tree of products: the finished good at the root, direct components as children, raw materials at the leaves. In a physical supply chain each product is made at some facility, and each component travels an inbound route to the point of use of its parent. The assignment of sites and inbound routes is the *sourcing* data of the BOM.

Abstractly, both the family of BOMs and the shipping network may be presented as directed containers. Sourcing then becomes a morphism of that structure—more precisely, a covariant functor of directed containers from the BOM container to the opposite shipping network. The content of this note is to make that functor precise, and to record the formalisation choice that makes it total.

The formalisation hypothesis. If the shapes of the BOM container were merely product-labelled trees, with no route data, a functor to the opposite network would have to produce an inbound path for every tree edge—including perverse trees whose edges run against the shipping graph. No such path need exist, so the data cannot extend to a total functor on all trees.

The remedy adopted here is to place the routing choices on the shapes themselves. A *routed BOM* is a product-labelled rooted tree indexed by the facility producing its root, in which every edge already carries an inbound route. On this domain the sourcing functor is canonical: no side data, and the functoriality equations hold definitionally (or by a single path induction for consolidation). The paper's informal data—a site per product and one leg per BOM edge—is exactly what builds a concrete routed BOM.

What this note claims. In ordinary mathematical language: routed BOMs form a directed container whose positions are occurrences (descending paths from the root). The shipping digraph G, read oppositely, forms a directed container whose positions at a vertex are inbound routes. Sourcing is the functor that sends a routed BOM to its root facility and an occurrence to the composite inbound route of that component. Functoriality is consolidation of multi-level routes. The UAE steel rebar example of the companion paper is a concrete instance, with the three-leg ore-to-rolling-mill route checked by computation.

We work throughout in a single universe of types, writing **Type** for that universe. Proofs are indicated by the phrase "by lean," meaning they have been checked by a proof assistant rather than written out in prose.

2 Background: digraphs, paths, and opposite networks

Definition 2.1 (Digraph)**.** A **digraph** G consists of a type V_G of vertices, a type E_G of edges, and source and target maps

$$\mathsf{src}_G,\ \mathsf{tgt}_G\ :\ E_G\ \longrightarrow\ V_G.$$

Definition 2.2 (Path)**.** A **path** in G from u to v is a finite walk of edges with matching endpoints. Paths form a category: there is an empty path nil_u at each vertex, and concatenation

$$\mathsf{append}\ :\ \mathsf{Path}_G(u,v)\times\mathsf{Path}_G(v,w)\ \longrightarrow\ \mathsf{Path}_G(u,w)$$

is associative, with nil as two-sided unit.

Definition 2.3 (Opposite digraph)**.** The **opposite digraph** G^{op} has the same vertices and edges as G, with source and target swapped:

$$\mathsf{src}_{G^{\mathrm{op}}}=\mathsf{tgt}_G,\qquad \mathsf{tgt}_{G^{\mathrm{op}}}=\mathsf{src}_G.$$

Consequently a path of G^{op} from v to u is an *inbound route* of G ending at v and originating at u.

Remark 2.4. The opposite network is taken on the graph, not as a general dual of directed containers: one forms the directed container associated to G^{op} in the usual way (shapes are vertices; positions at v are paths of G^{op} starting at v). Positions at a facility are therefore inbound routes to that facility, read backwards from the point of use.

3 Routed bills of materials

Fix a type P of products and a digraph G of shipping facilities and legs.

Definition 3.1 (Routed BOM)**.** A **routed BOM** over P and G, indexed by a facility $v \in V_G$, is a finite tree

$$t \;=\; \mathsf{node}(\mathit{prod},\; v,\; n,\; (v_i)_{i<n},\; (\rho_i)_{i<n},\; (t_i)_{i<n})$$

where

1. $\mathit{prod} \in P$ is the product label of the root,
2. $n \in \mathbb{N}$ is the number of direct components,
3. $v_i \in V_G$ is the facility producing the i-th component,
4. ρ_i is a path of G^{op} from v to v_i (the inbound route of that component into the root facility),
5. t_i is a routed BOM at v_i (the component's own BOM).

Raw materials are nodes with $n = 0$. We write $\mathsf{Bom}(P, G)(v)$ for the type of routed BOMs at v.

Thus a routed BOM is a product-labelled rooted tree whose every edge already carries its sourcing route. The routes live on the shapes so that the sourcing functor below is total.

Definition 3.2 (Occurrence)**.** An **occurrence** in a routed BOM t is a descending path from the root, including the empty path:

- here is the empty occurrence (the root itself);
- if $i < n$ and p is an occurrence in the i-th child, then $\mathsf{inside}(i, p)$ is an occurrence in the parent.

We write $\mathsf{Occ}(t)$ for the type of occurrences in t. Labels, sites, and routes play no role in the navigation.

Definition 3.3 (Sub-BOM and shift)**.**

1. The **sub-BOM** at an occurrence $p \in \mathsf{Occ}(t)$ is the subtree $t.\mathsf{sub}\, p$, together with the facility producing it.
2. The **shift** (occurrence concatenation)

$$t.\mathsf{shift} \;:\; (p : \mathsf{Occ}(t)) \;\longrightarrow\; \mathsf{Occ}(t.\mathsf{sub}\, p) \;\longrightarrow\; \mathsf{Occ}(t)$$

views an occurrence inside a component as an occurrence of the finished product.

Proposition 3.4 (Navigation laws)**.** *For every routed BOM t and occurrences p, q, r of the appropriate types:*

(D3) $t.\mathsf{shift}\, p\, \mathsf{here} = p$*;*

(D4) $t.\mathsf{sub}\, (t.\mathsf{shift}\, p\, q) = (t.\mathsf{sub}\, p).\mathsf{sub}\, q$*;*

(D5) *shift is associative along the transport of* (D4)*.*

Proof. By lean. □

Remark 3.5. Law (D4) is the slogan of multi-level BOM explosion: sub-BOMs nest along the shift, so multi-level costing is coherent. Law (D5) is the associativity of that nesting.

4 The directed container of routed BOMs

Definition 4.1 (BOM container)**.** The **directed container of routed BOMs** $\mathsf{bom}(P, G)$ has

1. shapes (v, t) with $v \in V_G$ and $t \in \mathsf{Bom}(P, G)(v)$;
2. positions of (v, t) equal to $\mathsf{Occ}(t)$;

3. root position here;

4. sub-shape and shift as in Definition 3.3.

The five directed-container axioms hold by the navigation laws of Proposition 3.4 (together with the definitional equalities for root).

Remark 4.2. Labels, sites, and routes ride along on the shapes without disturbing the navigation laws. The combinatorial skeleton is that of rooted trees; the supply-chain decoration is passive data.

5 The sourcing functor

Definition 5.1 (Inbound route of an occurrence)**.** Let t be a routed BOM at facility v. The **inbound route** of an occurrence $p \in \mathsf{Occ}(t)$ is the path of G^{op} from v to the facility producing the sub-BOM at p, defined by induction on p:

$$t.\mathsf{routeOf}(\mathsf{here}) \;=\; \mathsf{nil}_v, \tag{1}$$

$$t.\mathsf{routeOf}(\mathsf{inside}(i,p)) \;=\; \rho_i +\!\!+ \left((t_i).\mathsf{routeOf}\, p\right), \tag{2}$$

where ρ_i is the inbound leg stored on the i-th edge and $+\!\!+$ denotes path concatenation. Thus the empty occurrence is sourced by the empty route, and a nested occurrence composes leg-by-leg through intermediate producers.

Theorem 5.2 (Consolidation)**.** *For every routed BOM t and occurrences p, q of the appropriate types, the route of a concatenated occurrence is the concatenation of the routes:*

$$\left\langle (t.\mathsf{sub}\,(t.\mathsf{shift}\,p\,q)).1,\; t.\mathsf{routeOf}\,(t.\mathsf{shift}\,p\,q)\right\rangle \;=\; \left\langle ((t.\mathsf{sub}\,p).\mathsf{sub}\,q).1,\; (t.\mathsf{routeOf}\,p) +\!\!+ ((t.\mathsf{sub}\,p).\mathsf{routeOf}\,q)\right\rangle, \tag{3}$$

as positions of the opposite network (endpoint paired with route).

Proof. By lean. □

Remark 5.3. The two endpoints agree by (D4), but not definitionally; packaging them with their routes as dependent pairs lets the equality be stated without transport. Consolidation is the position-level form of multi-level sourcing: ore reaches the rolling mill only via the reduction plant and the melt shop.

Definition 5.4 (Sourcing functor)**.** The **sourcing functor**

$$\mathsf{site}_{P,G} \;:\; \mathsf{bom}(P,G) \;\longrightarrow\; \mathsf{network}(G^{\mathrm{op}})$$

is the covariant functor of directed containers given by

1. on shapes: $(v,t) \mapsto v$ (the facility producing the root product);

2. on positions: $p \mapsto \left\langle (t.\mathsf{sub}\,p).1,\; t.\mathsf{routeOf}\,p\right\rangle$ (the inbound route of the occurrence, ending at the component's own producing facility).

Theorem 5.5 (Functoriality)**.** *The maps of Definition 5.4 satisfy the three axioms of a covariant functor of directed containers:*

(F0) *sub-shapes are preserved (the network's sub is endpoint projection);*

(F1) *the root occurrence is sent to the empty route;*

(F2) *shift is preserved — this is consolidation (Theorem 5.2).*

Proof. By lean. □

Remark 5.6. Axioms (F0) and (F1) hold definitionally. Axiom (F2) is the single non-trivial content; its transport vanishes because (F0) is definitional. There is no side data: once a BOM is routed, sourcing is canonical.

6 Example: UAE steel rebar

We record the concrete instance that motivates the construction: a gas-based mini-mill making rebar for export.

Example 6.1 (Steel network and BOM)**.** The shipping graph is the directed path of four facilities

$$\text{ore berth} \longrightarrow \text{reduction plant} \longrightarrow \text{melt shop} \longrightarrow \text{rolling mill},$$

with three legs $\ell_{\mathrm{ob}\to\mathrm{rp}}$, $\ell_{\mathrm{rp}\to\mathrm{ms}}$, $\ell_{\mathrm{ms}\to\mathrm{rm}}$.

The products are ore pellets, DRI, scrap, billet, and rebar. The routed BOM at the rolling mill is:

- rebar at the rolling mill, with billets inbound along $\ell_{\mathrm{ms}\to\mathrm{rm}}$;
- billets at the melt shop, with DRI inbound along $\ell_{\mathrm{rp}\to\mathrm{ms}}$ and scrap already on site (empty inbound route);
- DRI at the reduction plant, with ore pellets inbound along $\ell_{\mathrm{ob}\to\mathrm{rp}}$;
- ore pellets at the ore berth, and scrap at the melt shop, as leaves.

Proposition 6.2 (Sourcing on the steel instance)**.** *Let t be the rebar BOM of Example 6.1, viewed as a shape of $\mathsf{bom}(P, G)$.*

1. *On shapes, $\mathsf{site}(t)$ is the rolling mill.*
2. *The ore-pellet occurrence is sourced by the three-leg inbound route*

$$\textit{rolling mill} \leftarrow \textit{melt shop} \leftarrow \textit{reduction plant} \leftarrow \textit{ore berth}.$$

3. *The scrap occurrence is sourced by the one-leg route ending at the melt shop.*
4. *Consolidation holds on the instance: sourcing the concatenated occurrence "billet then ore" equals shifting the billet's route by the ore's route in the opposite network.*

Proof. By lean. □

7 Discussion

What has been established. Routed BOMs form a directed container; the opposite shipping network is a directed container; sourcing is a covariant functor between them whose position action is multi-level route composition. The UAE steel example confirms the equations by direct computation.

Why routes live on shapes. The informal sourcing data of a companion development—a site per product and one inbound leg per BOM edge—cannot extend to a total functor on *all* product-labelled trees, because some trees demand routes that do not exist in G. Equipping each shape with its routing choices restores totality and yields a canonical functor. An alternative (restricting to trees compatible with a fixed component relation) makes shape equalities propositional and reintroduces transports throughout.

Scope. The development treats a single universe of types and does not address costing, lot sizes, or capacity constraints. Directed-container structure records only the combinatorial skeleton of navigation and sourcing; the economic content rides along as passive data on shapes and positions.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant; we have not reproduced the formal proof scripts here.

Acknowledgements

The mathematical content follows the theory of directed containers and the supply-chain reading developed in the companion notes on supply chains as directed containers. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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